



Analyse digital forensic evidences through a semantic-based methodology and NLP techniques

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HIGHLIGHTS

- The paper presents a semantic-based methodology to represent specific domain concepts.
- Authors choose the Computer Forensics domain to face-off Correlation problems.
- The reasoning mechanism derives implicit knowledge from extracted evidences.
- Authors also make use of NLP techniques to extract information from textual data.

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ABSTRACT

The increasing adoption of digital technologies to manage and process information used in everyday life, results in an increase in the demand for digital data analysis for investigative purposes. In fact, the reconstruction of computer and telematic crimes, or, in general, of crimes committed with computer systems, require the adoption of Computer Forensics best practices in order to extract relevant evidences from electronic devices, guaranteeing the integrity of data and their admissibility during a trial. The process of extraction, conservation, analysis and documentation of a forensic investigation can be enhanced by a framework that support investigators during their work, correlating evidences collected by different forensic tools. So, in this work we propose a semantic methodology and a system architecture for evidences correlation aiming to provide enhanced retrieval and reasoning capabilities.

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1. Introduction

Cybercrime is the term used by the Anglo-Saxons to indicate crimes involving computer systems; more precisely, it is possible to identify two categories related to cybercrime:

1. crimes for which IT or telematic systems¹ are the object of the offence²;

2. crimes for which information or telecommunications systems are the tools of criminal action; more precisely, the technological tool is used to “deceive” the user³ or to “facilitate” the commission of an offence.⁴

In last years there has been a real explosion of computer crimes [1], although the matter is still difficult for many legal operators. Since Information Technology (IT) systems are involved in almost all daily activities, it is inevitable that people who continuously use them leave fingerprints, just like they do in the real world. These digital fingerprints may range from web browser caches, to deleted file fragments, email, document, process logs, and backup files.

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¹ The term “computer or electronic system” is used to indicate in general any kind of computer tool (computers, mobile phones, storage devices, washing machines ...). In the Budapest Convention, art. 1 computer system is defined as “any equipment or group of interconnected or connected devices, one or more of which, on the basis of a program, perform the automatic processing of data”.

² An example of a crime for which the offence is a computer or electronic system is the infection with viruses or worms

³ An example is the phishing, a phenomenon of social engineering that, through the sending of unknown scammers by fraudulent emails, encourages victims to voluntarily provide personal information, such as credit card data.

⁴ An example is the case of the copying of confidential data protected by intellectual property from the company servers to carry out unfair competition practices

These virtual or digital traces can be valuable for any number of reasons. They may be useful as evidence in establishing the origins of a document or piece of software, for legal purposes in determining the activities of the parties involved in a criminal case. Moreover, the growing use of digital technologies increases the chances that a device should be the means or the witness of a crime. Thus, we can use data generated, stored or transmitted by a digital device as evidences in a trial, thanks to the fact that an information system is a mere container of digital data.

A research conducted by IISFA⁵ in 2012 has shown that more than half of the computer crimes pertain to abusive access to an information system, corporate infidelity (which often results in abusive access to an IT system in order to steal industrial secrets) and offences of child pornography committed through the use of file sharing clients on a peer-to-peer network [2].

However, the reconstruction of IT crimes is challenging, because it is difficult to state how many authors have been involved, where they acted, how many times an action has been committed and who are the victims [3]. Moreover the techniques adopted during the investigation of forensic evidences have to be compliant with the current regulation, standards and best practices, in order to guarantee the preservation of the computer evidence and the accuracy of the results regarding its elaboration [4].

Unlike *Computer Security*, that deals with the protection of an IT system, Computer Forensics intervenes after the violation has occurred or, in general, that the system has been actively or passively involved in a crime; the aim is the examination and documentation of the data contained within the informatics findings to reconstruct the events that have occurred [5]: a detailed analysis allows to know the activities, tastes, thoughts of the user in order to conduct the investigations in the right direction and acquire evidences related to the life of user.

2. Motivation

The first complete definition in Italian literature was made by Maioli [5] that qualifies the Digital Forensic science as “*the discipline that studies the set of activities that are aimed at analysing and solving cases related to cyber-crime, including the crimes realized with the use of a computer, directed to a computer or in which the computer can still represent a source of proof*”.

The affirmation of Computer Forensics established over the years in has been progressive and constant way, constant, following the evolution of technological devices. The growing number of different kind of devices caused the born of specialized branches, such as *Mobile forensics*, concerning the investigation of mobile devices, like tablets, GPS or, overall, smartphones. The last ones, thanks to their mode of operation, sum up the characteristics and the problems of all the other digital devices and systems, and, in relation to their diffusion, now far superior to that of computers, are affirming as a reference compendium of all the Computer Forensics problems.

Moreover, it should be borne in mind that, if once the data were saved exclusively in digital data storage devices connected to the computer physically used by the end user, today a lot of information is de-localized from a single location and stored in a network. Therefore, if a crime is committed even partially

through the use of a network, the tests are distributed on different computers and in many cases it is not physically possible to simultaneously collect all the hardware. At the same time, the presence of a network can induce redundancy, a phenomenon for which a data not available at one point in the network could be traced to another.

Taking into account the previous considerations, experts have to face many challenges during the different phases of forensic investigations:

- avoid data alteration to guarantee the admissibility of acquired data;
- extract and manage big volumes of data;
- integrate heterogeneous informations coming from different sources;
- manage and correlate data collected in different format files produced by acquisition tools.

2.1. Digital evidence

Digital evidence is the main element of any forensic process. Data is elementary facts, encoded information that needs an interpretation to take on meaning and provide knowledge. Digital data is a representation in a binary system of sequences of bits not immediately comprehensible to humans, so it requires a series of operations through which a transformation is performed that can lead to different results (shown on the monitor in textual representation or as a video, but also as an image printed on a piece of paper).

By its nature, digital data is:

- immaterial, for which it needs a suitable support to contain it such as CD-ROM, hard disk, USB sticks;
- volatile, as it can be dispersed quite easily;
- alterable, modifiable even in an anonymous and/or involuntary manner;
- reproducible in a potentially infinite number of copies.

Digital evidence can be considered any data allocated to a particular device or transmitted by IT and telematic systems that may have some procedural relevance [6]. Any data used to support or refute a thesis in order to define how an offence has occurred, or to establish the intention or the alibi is to be called scientific evidence in digital format. According to ISO/IEC 27037:2012,⁶ a digital evidence is every kind of information that can be stored or transmitted in a digital form and that can be considered an evidence.

Literature offers many definitions of digital evidence, such as:

- any data stored or transmitted using a computer that support or refute a theory of how an offence occurred or that address critical elements of the offence such as intent or alibi [7];
- information of probative value stored or transmitted in digital form [8];
- any digital data that contain reliable information that supports or refutes a hypothesis about an incident [9].

So, due to the easy alterable and destructible nature of digital evidences, care must be taken, even by the same investigators if not properly prepared to carry out a survey of computer forensics, not to compromise and contaminate the *scena criminis* status [10].

⁵ The IISFA (International Information System Forensics Association) is an association of professionals in the field of computer forensics (lawyers, law enforcement agencies and technicians) internationally recognized (<http://www.iisfa.org/>), whose primary purpose is the promotion of the study, the formulation of methods and standards concerning the Computer Forensics activities. In this regard, the Italian section of the association (<http://www.iisfa.net/>) is particularly active, organizing numerous refresher courses and seminars for both lawyers and technicians: among them the annual ISISFA event stands out. Forum, during which the latest results of studies on the topic are presented.

⁶ See Section 3.

2.2. The “Repeatability” of the investigations

The relation between bits and the medium on which they are stored rises a question of considerable importance for the definition of the technical methods and the rules for the correct processing of data for legal and procedural purposes. In fact, there are some considerations that must always be kept in mind for a correct setting of the method of data processing for procedural purposes:

- need for a memory (hard disk, floppy disk, flash memory): in cases where the bits are in the static phase, their physical size requires that you cannot assume documents that for the usability do not need a memory; when instead they are in a dynamic phase, the correct formation of the information deriving from the bits imposes that all the bits (or most of them) that are part of the flow are processed or acquired;
- reproducibility in infinite number of copies: thanks to their digital coding, the strings of bits can be replicated equal to themselves in an infinite number of strings always perfectly equal to each other, as verifiable by calculation and comparison of the respective hash.⁷
- volatility: the technology on which primary memories are based, also called volatile memories, requires continuous power supply, so that the bits processed by them are not permanently archived, but are lost if the power supply is stop;
- deterioration: the bits stored on the secondary memories only apparently seem to be destined to remain available for a long time; in reality, the ability to remain legible over time depends on the technology used for the construction of secondary memories. On a strictly physical plane, the secondary memories are affected by the intrinsic limits of the material which they are composed. These parameters condition the degradation of the bits and therefore of the data and of the information that can be deduced, so the bits stored on them are subject to deterioration and illegibility phenomena that go under the name of “digital obsolescence”. This phenomenon manifests itself for all digitized documents, until they are unusable;
- (almost) anonymous modifiability: by virtue of the mediation of the hardware and software in the process of creating and storing bits, they do not have physical elements that make it possible to trace unequivocally the operator who created, modified, transferred or deleted them, or to trace the characteristics that establish a relationship between bit and operator, a relationship that can be inferred on the basis of information that can be deduced from the specific physical characteristics of the bits.

These considerations not only constitute the minimal ontological and physical basis for a correct juridical qualification of the procedural facts relevant to the matter, but allow the following considerations to be made regarding the best techniques for the acquisition of the bits and therefore of the data both in the static phase (Computer forensics) and dynamic phase (Network forensics).

The findings also confirmed one of the postulates of computer forensics, i.e. the need to preserve the integrity of the bits that make up the data where it is intended to draw information to be used during the trial. The need to apply all the Computer forensic principles in processing data for trial use have as sole purpose preserving the integrity of the sequences of bits in order

to guarantee that the information derived can be reliable for all the parties.

With the expression *Repeatability* of the investigations we mean the possibility to completely re-run the analyses on models absolutely identical to the original; therefore repeatability is configured as a key factor, especially when working on crimes, and results must be presented in trial for an indictment. In other words, we assume the possibility of carrying out a post-mortem study of the systems, completely limiting the interactions with the outside world. The digital evidence sources validly obtained from digital forensics procedures, from the technical-scientific point of view, should possess the following characteristics:

- Integrity: it is the possibility of freezing the evidence found so as not to allow alterability over time. Once the source has been collected, this procedure entails the proper preservation of the data;
- Consistency: it consists in demonstrating that the information taken during the forensics analyses highlight facts logically correlated with each other in a temporal sequence that is coherent with the other procedural information also coming out of the digital environment;
- Documentation: the availability of the documentation details and certifies the adopted procedures, describing all the steps involved in the extraction and identification of the sources of evidence, including the methods and procedures used, the problems encountered, the ad-hoc solutions adopted, the tools with related version and license, the personnel who conducted the analysis with the relevant scientific qualifications, the chain of custody of the finds from the crime scene to the laboratory, etc.

3. Best practices and regulation

In regulating the *modus operandi* of the operations, it can be observed how the attention of the legislator has focused, rightly, more on the result that must be obtained rather than on the method to be followed: the canonization within legal norms of technical procedures more that to represent a guarantee, would have led, in the long run, to contrary and distorting effects represented by the constant evolution of the discipline and by the peculiarities peculiar to each case.

Until October 2012 the methodologies were defined in some best practices of the sector, aimed at outlining the paradigms of technical action in the forensic field, through a basic methodology aimed at: (a) acquiring the test without altering or damaging the device original; (b) authentication of the find and image (bit stream image) acquired; (c) to guarantee the repeatability of the assessment; (d) an analysis without modification of the original data; (e) maximum impartiality in technical action.

Among the various standards promoted by the ISO, documents relating to Computer Forensics have recently been submitted, which are candidates for effective and internationally recognized technical standards of reference:

- ISO/IEC 27037:2012,⁸ issued in definitive version on 15 October 2012, define the guidelines that have to be followed in any investigation that needs to maintain the integrity of digital evidence;

⁸ It refers to other ISO/IEC standards:

ISO/TR 15801 – Document management – Information stored electronically – Recommendations for trustworthiness and reliability.

ISO/IEC 17020 – Conformity assessment – Requirements for the operation of various types of bodies performing inspection.

ISO/IEC 17025:2005 – General requirements for the competence of testing and calibration laboratories.

ISO/IEC 27000 – Information technology – Security techniques – Information security management systems – Overview and vocabulary.

⁷ <https://it.wikipedia.org/wiki/Hash>.

- ISO/IEC 27041, with regard to guidelines on the guarantee of suitability and adequacy of the methods of investigation;
- ISO/IEC 27042, related to guidelines for the analysis and interpretation of digital evidence;
- ISO/IEC 27043, concerning principles and processes for the investigation of IT incidents.

4. Methodology's phases

Taking into account the various definitions of computer forensics, the state of technology, various guidelines defined in the best practices and, lastly, the ISO/IEC standards on the topic, the process of computer science forensic could be divided into seven phases: identification, collection, acquisition, conservation, analysis, evaluation and presentation.

4.1. Identification

The identification phase consists in searching evidences useful to the case. The digital evidence has a physical form and a logical form: the physical form is given by the technique of impression of data on the support (for example, magnetization of matter in the case of magnetic supports or the representation in pit and land of optical media); the logical form is the virtual representation of the data in bits that can take the value of 0 or 1. Identification is the process of research, recognition and documentation of potential evidences in digital format, however, this is not just a simple search for devices, but the right priorities must be defined (taking into account the risk of data volatility) in order to minimize the damage to potential digital evidence and obtain the most intact and genuine data [11].

4.2. Collection

The collection phase consists in obtaining “materially” the data to be analysed. The most common findings are computers, which very often are not handled correctly during the seizure phase, not considering the fact that they have easily alterable storage media.

If a computer is found turned on, it is advisable to make some considerations before proceeding further. First of all, it is essential to understand what data to expect compared to the considered crime. In some cases it may be necessary to make a copy of the RAM memory, which loses its contents at shutdown, to ascertain which programs were running at the precise moment of the seizure. Then, it is necessary to evaluate the best shutdown mode, between standard shutdown (shutdown) and the disconnection from the power supply (unplugging). The former is usually inadvisable because it causes numerous modifications to the system files, but the latter could be fatal for old or delicate hardware, with the risk to make the machine unusable.

The forms through which collection can be implemented are the duplication or the seizure. Usually the form preferred by law enforcement is the seizure, whose critical point is only the care of the physical support and the correct maintenance of the chain of custody. However, it has some advantages:

- **simplicity and tranquillity:** the collection of the physical medium does not require the particular technical knowledge necessary for the acquisition, although the removal of digital storage media still requires some skills; moreover, the postponement of the acquisition operation helps to loosen the tension in critical moments of a seizure activity, avoiding errors;
- **rapidity:** the collection simply requires the annotation of the identification details of the support in a report, in addition to the workload required for transport;

- **tangibility:** the tangibility of the support conveys greater peace of mind to the operators and to the suspect who may not have adequate technical and legal knowledge to evaluate the current technical activity;
- **preservation of further non-digital tests:** in addition to the digital data, an IT evidence could be used to detect other types of tests such as fingerprints.

The alternative procedure, is the duplication of the medium through bit-to-bit copies, which can be necessary under different circumstances where physical collection is not possible:

- **IT systems that cannot be turned off:** these are systems that deliver critical services in 24/7 mode; for example, systems for controlling the exchange of railway tracks;
- **IT systems that provide services to third parties:** these are systems that typically located in data-centres and provide resources, both computational and storage, to various users, allowing them to reduce costs by centralizing the investment of hardware and software, as well as the costs for the system activity; for example, systems of hosting service providers that host websites;
- **virtual systems:** these are systems that simulate a real machine whose physical consistency is that of the system on which the activity is performed.

In any case, the collection process must not be limited to the device that contains the digital data but must be extended to the material that concerns it, such as, for example, post-it with passwords, notebooks with notes and power supplies. Furthermore, the collection process must be documented in detail to justify the choice of one method instead of another. The IT media must be carefully packaged away from heat sources so as not to risk corrupting the media and causing accidental data loss.

The following scheme in Fig. 1, taken from the ISO/IEC 27037:2012 standard, highlights the evaluation process by the investigator, regarding the possibility of making a collection or acquisition. Based on the choice made, the same standard provides detailed procedures to follow.

If you were to tend to collect, in the case of devices turned off the scheme in Fig. 2a is to follow. Instead, when you come into contact with a turned on device, the scheme, reported in Fig. 2b, is more complex because it requires the evaluation of some elements that would be lost permanently after switching off the system. Note that after switching off the system (with unplugging or with normal shutdown), the procedure is based on that of the previous diagram for the devices switched off.

4.3. Acquisition

Acquisition is the process of producing a copy of digital evidence called a forensic copy (or bit-stream image, or a bitmap image), which is a complete copy of the media including unallocated space and slack space.⁹ The operating methods and tools to be used depend on the situation, cost and timing, but in any case must be documented in detail so that they are, where possible, reproducible and verifiable by a technical consultant of one of the other parties. The acquisition process must be as least invasive as possible, i.e. it must involve the alteration of the least possible number of bits, possibly aiming at the inalterability of the source support, in order to produce a sequence of bits representing the original sequence. The final product of an acquisition can

⁹ Slack space is a set of digital data generated by the way the data is organized into a storage medium. Regardless of the file size, media are structured in fixed-size blocks (sectors). Any unused space within the block will contain data that existed until the time of deletion and will continue to contain it until the space is overwritten as a result of wiping or new file allocation.

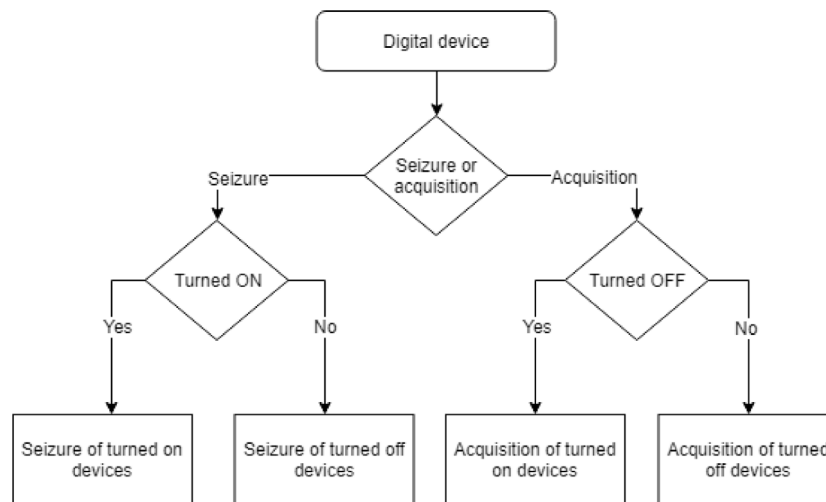


Fig. 1. Decision-making policy about the opportunity to collect or acquire potential digital evidence taken from the ISO/IEC 27037:2012 standard.

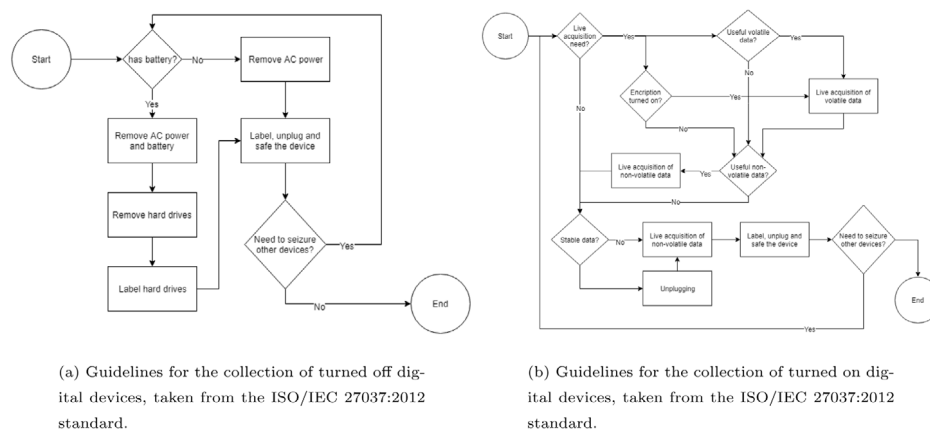


Fig. 2. Caption place holder.

therefore be a clone, i.e. a device that contains the exact and identical sequence of the source device, or an image file (or a series of fragmented files) that represent the exact and identical sequence of the device source; in the second case it is possible to apply compression algorithms as in the case of the Expert Witness format.¹⁰ The identity between source and destination can be easily proved by hash algorithms, mathematical functions that allow to synthesize the representation of millions of bits in hexadecimal strings of a few tens of bits: in fact, the application of a same hash function to two bit sequences always produces the same result (digest), if and only if the two input sequences are identical. It should be noted that in some circumstances – such as the live acquisition¹¹ of a system – this verification is not possible, therefore, it is necessary to resort to an in-depth documentation of the acquisition process, also by means of video and photographic recording.

If the machine is still on at the time of the seizure, it is necessary to carry out a first immediate analysis on site because the shutdown would result in the loss of volatile data in RAM memory or in the swap area and the inaccessibility of data protected by encryption. Live analysis must therefore verify network

connections, open ports, running process and their behaviour, current users, files, kernel modules and devices in use [12].

It can then be switched off and disconnected from the network to conduct *post-mortem* analysis, i.e. with the machine turned off, which consists of a survey of resident data (data files and programs), system file and logs analysis.

The investigations should not be carried out on the original hard disk but on a copy of the bit stream: the physical copy compared to the logical one allows to preserve data apparently not present on the disk that are instead highlighted with a forensic analysis.

4.4. Analysis

The analysis activity consists in recovering those data that may be useful for the purposes of a forensic investigation, which are therefore likely to be hidden, voluntarily or not, from the view of a common user.

ISO/IEC 27042:2015 – *Information technology – Security techniques – Guidelines for the analysis and interpretation of digital evidence* offers a guidance on the process of analysing and interpreting digital evidence.

The standard emphasizes the integrity of the analytical and interpretational processes, touching on issues such as the selection and use of forensic tools, plus proficiency and competency of the investigators.

¹⁰ EWF is a proprietary data format used for forensic copies devised by EnCase but now supported by most forensic software.

¹¹ The term *live forensics* indicates a method of acquiring information from an IT system while it is operating, in order to capture those data, transiting or stored in it, which would not be acquired if the device is turned off.

4.5. Valuation

A further significant aspect concerns the determination of the circumstances in which a crime is committed and the modalities of the same; even if the victim can be known, reconstructing details is essential to shed light on what happened.

It should be clarified that the reason for which an evaluation phase of the exhibit is necessary lies in the fact that the bit is 0 or 1, hence absolutely poor information. Since the information can be altered, polluted, counterfeited, it is necessary to ascertain whether these events occurred, if they were potentially verifiable and whoever could have done these actions. Another aspect concerns the acquisitions operations, for which it is necessary to assess whether they have been carried out with rigour and in a correct manner in compliance with the current legislation. Once these requisites have been assessed, judgements must therefore be formulated regarding the reliability of the informatics finding, in the sense of its integrity and verifying possible alterations, and its authenticity, ascertaining the author or the authors. So, during the evaluation phase a meaning is given to the data emerged during the analysis phase and the legitimacy of the operations carried out to acquire them is ascertained [5]. A valid judgement is issued regarding the reliability, integrity and authenticity of the find, carefully evaluating every possible event of alteration (who could, how and when).

4.6. Presentation

The last phase of the forensic examination consists in the presentation of all the evidence found by the consultant and his conclusions in a detailed report that will be considered during the trial. Within the technical report all the documentation acquired or produced during the analysis must be included. This means is essential to bring in the “knowledge” of the technical activities carried out during the investigation phase. The methodology used to analyse the data, the tools used, the discoveries made must be indicated in an exhaustive way, providing an explanation of what was done, why, by whom and how long each operation was performed.

The purpose of the presentation is to transmit to all the parts of the process the facts ascertained according to scientific techniques and methodologies, which will illustrate the steps taken. The report will be more effectively illustrated where there will also be audiovisual instruments (e.g. slides, recordings, etc.), simulations and practical demonstrations to support deposition.

5. Architecture of evidences analysis system

The wide range of investigation scenarios and the incessant technological evolution does not allow the definition of a unique procedure for the analysis of digital evidences.

Forensic tools play a major role in the acquisition phase, since they adapt acquisition processes to different digital devices (e.g. hard disks, USB flash drives, mobile devices, IDS Firewall logs, memory, etc.) [13–15]. Also social network analysis techniques are becoming crucial to investigate potential dangerous behaviours of the users [16]. The specialized and evidence-oriented design of forensic tools produces acquired data in different format and representations, that get more difficult the analysis process and so calls for advanced interoperability techniques for evidences correlation.

The development of automatic processes that assist detectives in data acquisition and analysis phases simplifies their work, especially when they have to deal with large amount of data. Currently, the most of forensic tools work on plain text data

Table 1
Number of events for each category of considered dataset.

Category	N. of events
Critical	42
Error	7344
Warning	1193
Information	29405
Total	37984

which does not allow advanced analysis processes. A semantic approach, eventually based on ontologies, use a unique and reliable representation of domain concepts, enabling data structuring on one hand, and standardized representation of data on the other hand.

Unlike structured data formats (i.e. relational databases), ontologies analysis tools easily make inference of new information, checking knowledge consistency, etc. This because an ontology explicitly represents relationships between entities.

The implementation of the system is adaptable to the needs of the specific domain and allow the reconfiguration of the storage, processing and networking resources. In this sense, Cloud Computing technologies [17] provide a suitable infrastructure in order to deploy the architectural model proposed in Fig. 3.

5.1. Evidences collection

A module called “Evidence manager” loads the binary content from the acquired data (i.e. the bit-stream image in the case of duplication, or from a dump or a filesystem in the case of live acquisition), identifying the type of current data and verifying its integrity through hash values. This module provides different tools to extract relevant artefacts from a forensic image, like user files, browser history, Windows registry, etc. The knowledge extracted consists of a set of attributes, including temporal information (date, time and time-zone), if the item was deleted, a description of the information source (source and source type), and the binary content of the data. Many data are already structured and ready to be used as instances attributes; instead, some of them, like description of the event, are non-structured fields and they require further processing based on regular expression or Natural Language Processing techniques.

5.2. Unstructured data processing

This phase is crucial for an effective analysis of all data stored on a media, and it allows to reduce the time and the human effort related to a visual inspection of retrieved information.

In our model, we foresee different processing resources, according to the type of data to be analysed: as already stated, forensic images can contain textual documents as well as multimedia information like pictures, video streaming and audio information. So the system is composed by a text processing module, and by a multimedia data processing module.

Text processing. The Text Processing module aims at extracting relevant information from text, associating specific concepts to the related key terms and defining relationships among them. The text is processed in according to the following: (1) Linguistic analysis: performs a morpho-syntactic analysis of the text (i.e., text tokenization and normalization, Part-of-Speech Tagging, lemmatization and complex terms analysis) combined with statistical analysis, thus enabling the extraction of relevant terms. These terms and the information about them, refined with the help of domain experts, will constitute a lexicon that is exploited for the building of the set of concepts used for the domain formalization via ontologies. (2) Semantic analysis: by using the information of

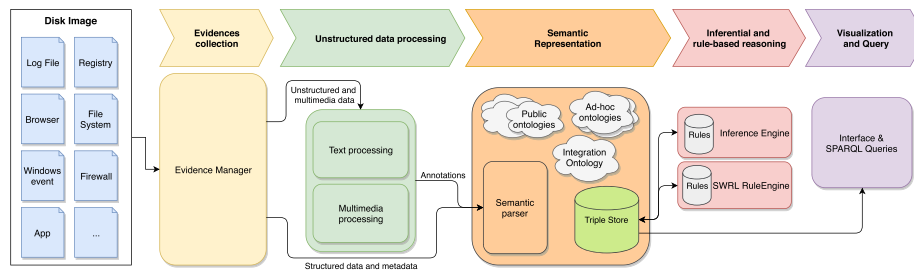


Fig. 3. Architecture's components.

the early analysis, it detects properties and associations among terms, defining the concepts and relationships, allowing ontology building and final documents annotation.

To implement this module we exploit the functionalities provided by GATE (General Architecture for Text Engineering). The platform is suitable for both “rule-based” and “machine learning” approaches and includes the following components:

- GATE Developer: an integrated environment for the development of language processing components (“language processing”) supported by a wide range of information and examples and enriched by a wide selection of additional components (“plugins”) that extend the potentiality;
- GATE Teamware is a collaborative web environment for the team management of document set annotation processes, based on the use of the JBPM “workflow engine”.
- GATE Embedded: a framework and a library of components for integration and use from an external application of services and processes implemented through GATE Developer;
- A “cloud” solution for processing “hosting” large-scale NLP /IE processes (GATE Cloud.net);
- GATE Mimir: (Multi-paradigm Information Management Index and Repository) used for indexing and searching for texts, annotations, semantic schemas (ontologies) and instance values. It allows you to perform composite searches through the free combination of textual, structural, linguistic and semantic data.

The development of a new application based on the architecture of GATE needs a preliminary phase, in which the developer chooses how many and which processing resources to use, defining what are the language resources involved for each of them. The main processing body consists of the pipeline, intended as a sequence of “process resources” (PR) each dedicated to a particular task in relation to the needs of NLP and Information Extraction. Developing a pipeline on GATE platform, through GATE Developer, requires to compose, according to the objectives and the type of processing, a set of selected PR, among those available within the plugins. For specific domains it may also be necessary to realize, in JAVA language, specific process templates, following the standards and indications of the GATE Embedded framework.

The output produced by a GATE application is a set of annotated data. The “annotation” in GATE represents a sequence of words that is the object of recognition, defined, in its breadth of coverage according to the identification and extraction objectives; has an appellation (“annotation type”) and properties (features) designed to contain derived values during the IE process. The naming of the “annotation types” and “features” are determined within the application according to the meaning and the requirements of formal representation. The annotations, as carriers of information (a name and properties), can be created through successive phases, increasing the detail of the representation being used in the “rule based” approach in a subsequent refinement up

to the determination of the final annotations of interest. At the end of the process the intermediate, service annotations, if no longer necessary, can be deleted.

An interesting application based on GATE is ANNIE (A Nearly-New Information Extraction system). ANNIE uses a series of resources to recognize and “annotate” instances of entities (people, organizations, places, dates, addresses, etc.) within a text. ANNIE is based on finite-state algorithms and the Java Annotation Pattern Engine (JAPE) language.

All the generated annotations can easily be recorded on DB or external files to be reprocessed and each generic pipeline can be executed by an external JAVA application through the interfaces exposed by GATE Embedded framework. GATE Developer also provides a plugin that allows the management of OWL ontologies in RDF/XML, N3, NTriples and Turtle formats, and therefore compatible with the similar tools available on other platforms.

It is evident that the organization of glossaries and grammars of rules is closely related to the context of application (“domain”), with respect to which particular terminologies, abbreviations, expressive forms and meanings they can involve significant and essential specializations in the different phases of the elaboration process.

Multimedia data processing. The Multimedia Data Processing module aims to classify multimedia objects, associating concepts from the domain ontology. This module exploits Artificial Intelligence techniques (such as Machine Learning, Convolutional Neural Network, etc.) and low-level features of the processed multimedia resource, in order to derive additional information from multimedia objects.

5.3. Semantic representation

Data coming from the previous phase constitute the input for a “Semantic Parser” module. Semantic parser module generates an OWL representation of knowledge extracted from digital evidence in the previous step. This module instantiates the ontology, combined from public domain ones, if available, or custom ones. Ad-hoc ontologies can be created to integrate all the referenced domain ontologies or to define new classes, additional restrictions, new object properties, etc. Also, ontology integration promotes reuse of entities defined in other ontologies and increases system flexibility, since domain ontologies may not be easily modifiable. For example, in Fig. 4 is reported the graph representation of the Computer Network ontology schema.

Ontology population is made by creating an instance (in the form of a triplet, subject–predicate–object) for each footprint item of acquired data, and by linking them each other according to formal object properties defined in the ontology schema, like in Fig. 5.

Final information is stored as RDF assertions into a Knowledge Base and all the knowledge associated to a document is in turn managed by proper ontology repositories.

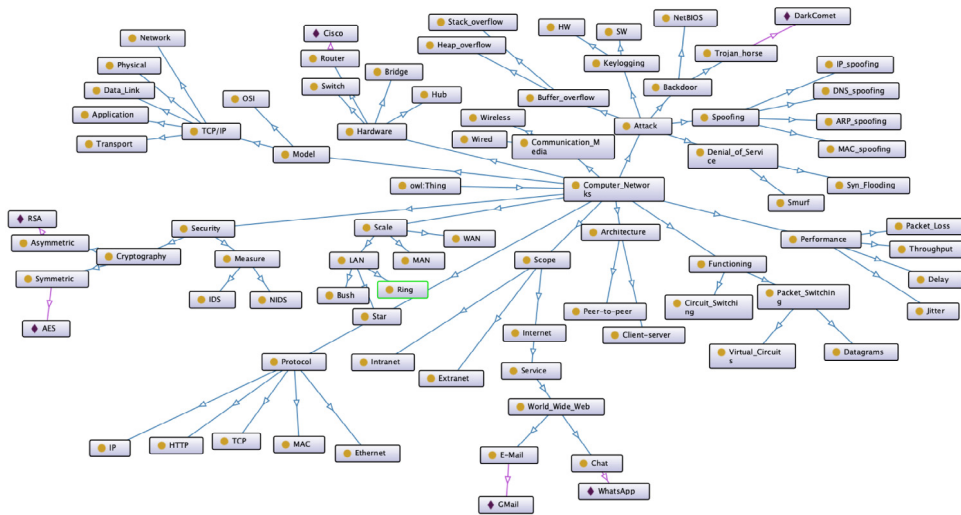
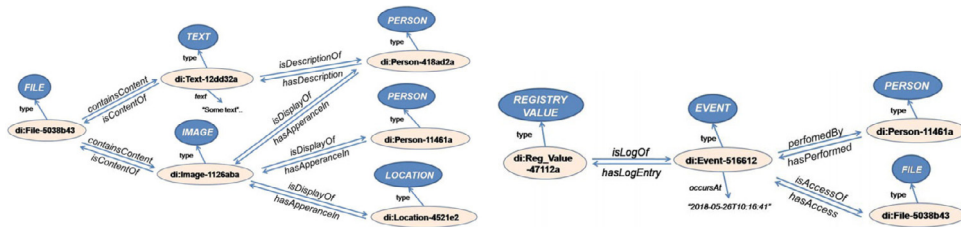


Fig. 4. Graph representation of Computer Network ontology schema.



(a) Example of instances' relations between File, Text, Image, Person and Location. (b) Example of instances' relations between Registry Value, Event, Person and File.

Fig. 5. Examples of ontology population.

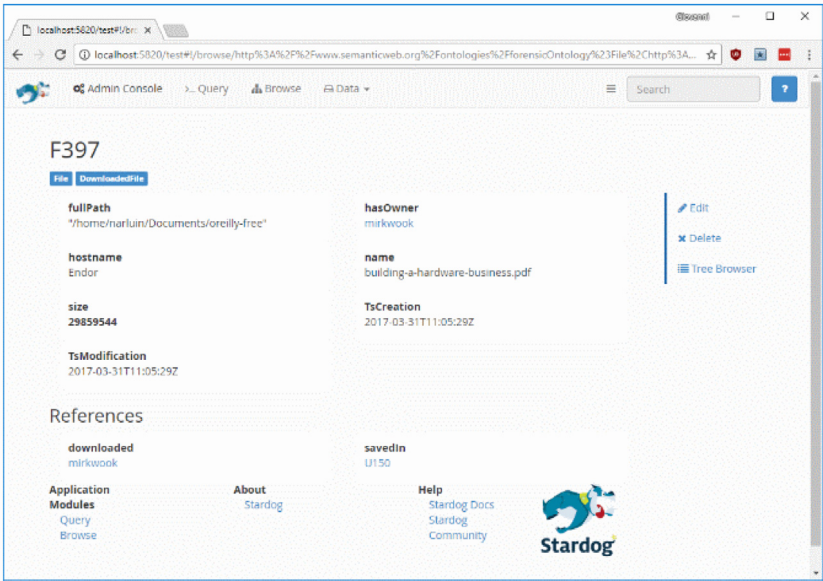


Fig. 6. Example of SPARQL query result.

5.4. Inferential and rule-based reasoning

An Inference Engine performs automated reasoning, according to the OWL specifications, coming from domain ontologies or investigators knowledge. Reasoners can specify the granularity of

inferences to be made, such as hierarchy relationship (an individual that is a member of a subclass is also a member of the parent class), or the generation of inverse object properties. Such kind of inferences can improve the performance of query execution. The goal of this step is to enrich the knowledge base with new inferred facts, increasing the examiner's knowledge on evidence.



Fig. 7. Numbers of detected, false positives and undetected events.

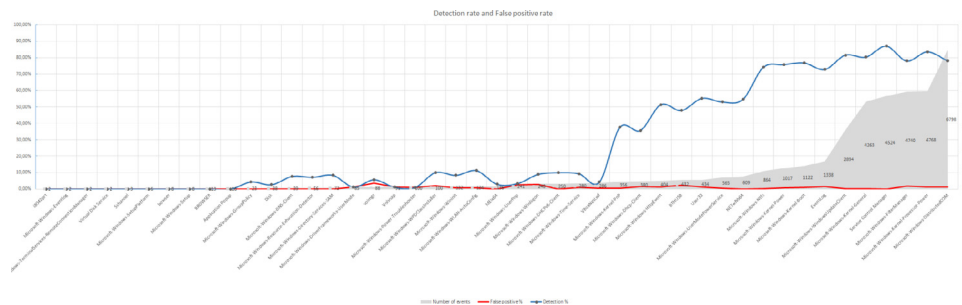


Fig. 8. Detection rates and False positive rates for considered dataset.

For example, it is possible to infer a file type from its extension or to deduce the author of an action from user information in its active account.

In addition, SWRL (Semantic Web Rule Language) rules play a major role in the automated integration and correlation parts of the methodology. SWRL Rule Engine adopts SWRL rules in order to correlate different individuals or to establish relationships among individuals belonging to different ontologies but representing similar concepts. This module can be called before the reasoning engine make it able to process the axioms that can only be inferred by the SWRL rules, or you may execute the reasoning step twice, one before the rule engine and the other after it. For example the Listing 1 reports an example of a SWRL rule inferencing the orphan connections through a set of network traffic records.

Listing 1: Example of SWRL rule.

```
PREFIX rule: <tag:stardog:api:rule:>
PREFIX : <urn:test:>
PREFIX f:<http://www.semanticweb.org/
ontologies/forensicOntology#>
IF {
  ?machine a f:Machine .
  ?access a f:Access .
  ?access f:IpAddress ?access_ip_address .
  ?machine f:IpAddress ?machine_ip_address .
  filter (?ip_address != machine_ip_address )
  .
}
THEN {
  ?access a f:Orphan .
}
```

the semantic parsing of the source data or inferred by the reasoning or the rule engine are loaded in-memory and SPARQL queries can be evaluated against it. An example of query is shown in Listing 2 and the related results are shown in Fig. 6. Once more, the queries can be saved in separate files promoting reuse and decoupling.

Listing 2: Example of SPARQL query.

```
PREFIX f:<http://www.semanticweb.org/
ontologies/forensicOntology#>
INSERT {
  ?u a f:Download .
  ?f a f:DownloadedFile .
  ?u f:savedIn ?f .
  ?user f:browsed ?u .
  ?user f:downloaded ?f .
}
WHERE {
  ?u a f:URL .
  ?u f:size ?su .
  ?u f:url ?url .
  ?f a f:File .
  ?f f:size fSize .
  ?f f:hasOwner ?user .
  ?f f:name ?fname .
  FILTER (?su = fSize)
  FILTER contains(?url,?fname)
}
```

6. Evaluation

A complete discussion about time and space complexities of the whole methodology is complex since they depend, during each step, on many, different, elements. During the first stage (collection), as previously described, we collect all digital evidences from the devices we are analysing. Hence, the elements

5.5. Visualization and query

The final module is responsible for accepting SPARQL queries from the user and evaluating them against a SPARQL query engine. The set of RDF triples that have been either asserted during

Table 2
Number of events for manually labelled events.

Manual annotation	Critical	Error	Warning	Information	Total	Detected
Application Popup				16	16	15
browser		8			8	7
BROWSER				13	13	12
BTHUSB			288	144	432	371
Disk		10	28		38	37
EventLog		24		1314	1338	1150
i8042prt			2		2	1
MEI{x64			2	134	136	127
Microsoft-Windows-Dhcp-Client				380	380	330
Microsoft-Windows-DHCPv6-Client				250	250	245
Microsoft-Windows-Directory-Services-SAM				72	72	70
Microsoft-Windows-DistributedCOM		6798			6798	6336
Microsoft-Windows-DNS-Client			39		39	35
Microsoft-Windows-DriverFrameworks-UserMode				85	85	76
Microsoft-Windows-Eventlog		2			2	1
Microsoft-Windows-FilterManager				4740	4740	4171
Microsoft-Windows-GroupPolicy				23	23	21
Microsoft-Windows-HttpEvent			400	4	404	379
Microsoft-Windows-Kernel-Boot				1122	1122	987
Microsoft-Windows-Kernel-General				4263	4263	4203
Microsoft-Windows-Kernel-PnP			356		356	302
Microsoft-Windows-Kernel-Power	42			975	1017	884
Microsoft-Windows-Kernel-Processor-Power		16		4752	4768	4338
Microsoft-Windows-Ntfs				864	864	838
Microsoft-Windows-Power-Troubleshooter				100	100	96
Microsoft-Windows-Resource-Exhaustion-Detector			56		56	54
Microsoft-Windows-Setup				8	8	7
Microsoft-Windows-SetupPlatform				6	6	5
Microsoft-Windows-TerminalServices						
-RemoteConnectionManager				2	2	1
Microsoft-Windows-Time-Service			18	262	280	260
Microsoft-Windows-UserModePowerService				565	565	542
Microsoft-Windows-UserPnp				241	241	204
Microsoft-Windows-WindowsUpdateClient		55		2839	2894	2807
Microsoft-Windows-Wininit				132	132	125
Microsoft-Windows-Winlogon				248	248	210
Microsoft-Windows-WLAN-AutoConfig			4	130	134	123
Microsoft-Windows-WPDClassInstaller				100	100	86
NETwNb64				609	609	609
Schannel		3			3	2
Service Control Manager		39		4485	4524	4524
User32				434	434	381
VBoxNetLwf		286			286	280
Virtual Disk Service				2	2	1
volmgr		88			88	74
Volsnap				91	91	81
Total	42	7344	1193	31102	39681	37105

that mainly influence time and space complexities are the number of elements to collect, i.e. the dimensions of file systems and of logs to analyse. In addition, the whole time for collection depends on access time too.

The complexity of all the other phases of the methodology depends on the ontology definition (in terms of how many classes, properties and relationship are defined in it): the more relationships are in the ontology, the more is the complexity of each step, because relationships need cross analysis. However, the number of generated triples depends also on the actual values of collected evidences, because each evidence can (or not) contain properties or relationship expected by the ontology definition.

We firstly conducted an experimental campaign to prove the effectiveness of the proposed methodology regarding the document analysis phase. In particular, we show that we can automatically discover many abnormal behaviours in a logging system application. To measure ad-hoc information retrieval effectiveness in the standard way, we used a test collection of logging file. Each of the sample documents must be manually annotated indicating, not only if it contains normal or abnormal event, but also the instances of interest, in our case the name of the event and the belonging category among those defined within our

system (Critical, Error, Warning, Information). [Table 1](#) shows the number of events of each category for the considered dataset.

In [Table 2](#) are reported the number of events for all the categories and the manual labels. In the last column is also reported the number of detected events through the presented methodology.

From the [Fig. 7](#) we can observe that the prototype system detects a large fraction of relevant documents within the considered dataset. In [Fig. 8](#) is clearly shown that for the most representative categories of events, those containing a great number of events, the detection rate is greater than 85%, while the reported rate of false positives is never greater than 3%.

After data collection phase, we focus on the evaluation of the ontological representation and enhancement (reasoning and rule evaluation) phases, considering two of the most representative categories of events in our test dataset. [Table 3](#) reports the number of extracted entries, the number of triples generated during the enhancement phase and finally the number of significant correlations found. [Table 4](#) gives the execution times (in seconds) for the different phases of the process.

From the [Table 4](#) we can derive that the analysis phase is the most time-consuming step. Moreover, we can see that the number of triples does not increase linearly with the number of

Table 3

Volumes of processed data through the reconstruction and analysis process.

Criterion	Dataset No1	Dataset No2
Extracted entries	6798	4263
Generated triples	8786	10475
Deduced triples	18	21
Correlations	1231	535

Table 4

Volumes of processed data through the reconstruction and analysis process.

Steps	Dataset No1	Dataset No2
Collection	1.87	1.43
Representation	24.37	18.3
Reasoning	0.29	0.247
Analysis	776.3	241.3

the acquired evidences. The variance of the results stresses the dependence of the methodology on the complexity of the input documents and logs.

7. Conclusions

In this work we presented an extensible methodology that exploits text processing techniques to represent relevant concepts of a specific domain in a standard and formalized format, according to Semantic Web principles.

We considered the Computer Forensic domain to face-off the Information Extraction and Retrieval challenges and to enhance the correlation of collected evidences. This approach intrinsically brings the advantages of making relationships between the domain entities and of the reasoning and inferential mechanism.

In future works we planned to make further development of the framework in order to improve the performances of analysis phase and the suitability of the extraction phase to a more wider range of devices and sources. Moreover, we are investigating the exploitation of machine Learning and multimedia analysis techniques, aiming to the production of more meaningful representation of multimedia resources, for example, by annotating an image with the name of the location or the name of the subjects depicted in it.

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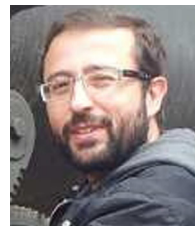
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